DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT

Calculus and Optimization

Functions and Graphs

Lecture No. 02





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Recap of previous lecture



Topic

FUNCTIONS & THEIR GRAPHS - 1

Topics to be Covered



Topic

FUNCTIONS & THEIR GRAPHS - 2

RECAP:

Function: if $\forall x \in A$ $\exists$ unique $y \in B$ s.t $y = f(x)$ then f is called func^n

(y: output, x: Input)

Domain: set of permissible Inputs for f is called Domain.

Range: " " outputs " " " Range. func^n.

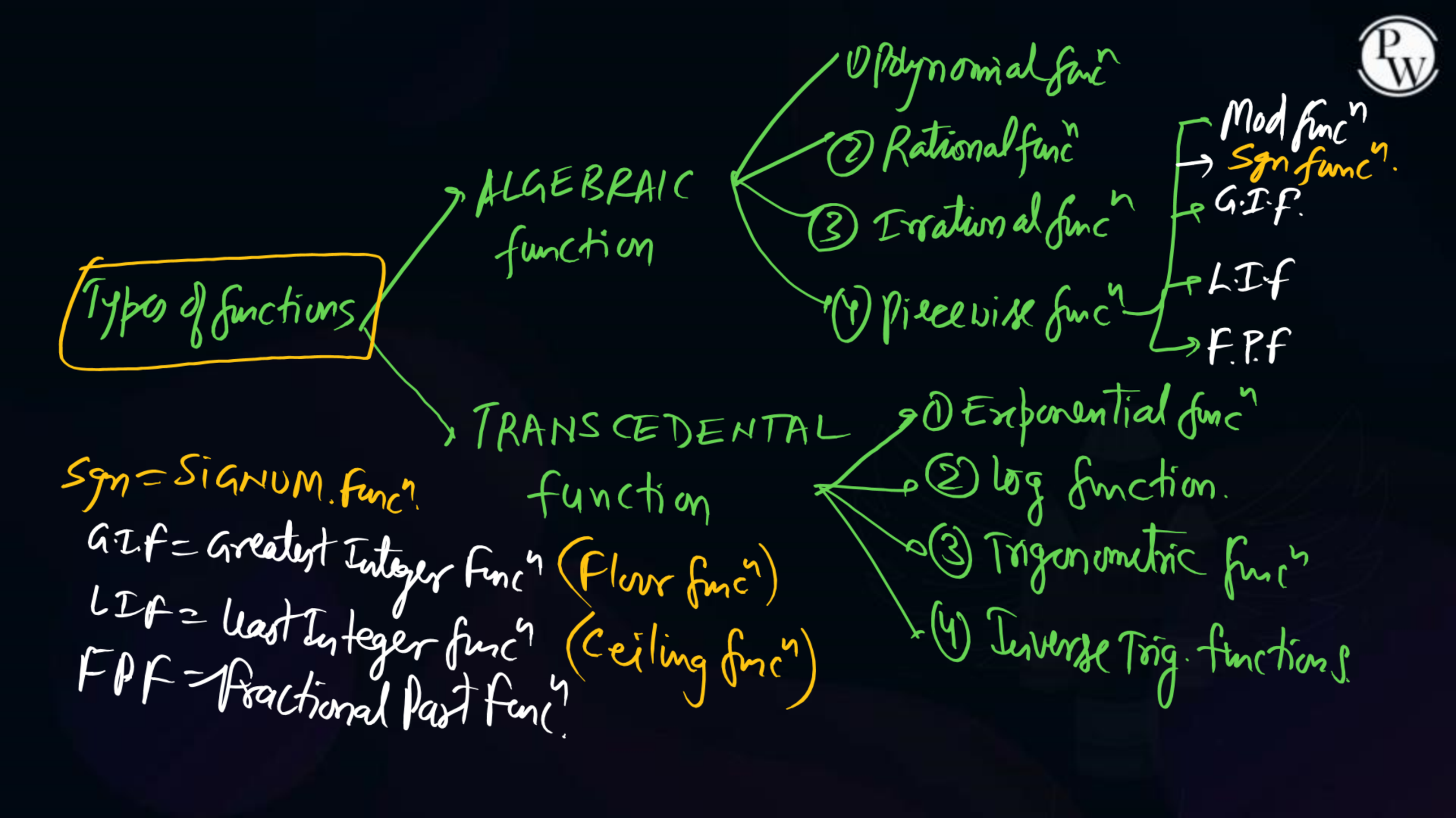
V-line Test → Line $\perp^r$ to y axis cuts the graph only at one point then it is a

Even func^n: if $f(-x) = f(x)$ Symmetry about y axis occurs

Odd func^n: if $f(-x) = -f(x)$ " " origin occurs

NEITHER func^n: if $f(-x) \neq f(x)$ & $\neq -f(x)$

Periodic func^n: if $f(x+T) = f(x)$ i.e it's graph Repeats after Period 'T'


Types of functions
ALGEBRAIC function
① Polynomial func^n
② Rational func^n
③ Irrational func^n
④ Piecewise func^n
Mod func^n
Sgn func^n.
G.I.f.
L.I.f
F.P.f
TRANSCEDENTAL function
① Exponential func^n
② log function.
③ Trigonometric func^n
④ Inverse Trig. functions
Sgn = SIGNUM. func^n.
G.I.f = Greatest Integer Func^n (Floor func^n)
L.I.f = Least Integer func^n (Ceiling func^n)
F.P.F = Fractional Part Func^n.

Polynomial: It's Domain is $(-\infty, \infty)$ & Degree $= 0, 1, 2, 3, 4, 5 \ldots$ ——.

& it's Definition is same at all points in the Domain of $y = f(x)$

eg $y = k$ (Constant poly) $\simeq$ degree $= 0$

$y = ax + b$ (Linear poly) $\simeq$ Deg $= 1$

$y = ax^2 + bx + c$ (Quad. poly) $\simeq$ Deg $= 2$

$y = ax^3 + bx^2 + cx + d$ (Cubic poly) $\simeq$ Dy $= 3$

⋮

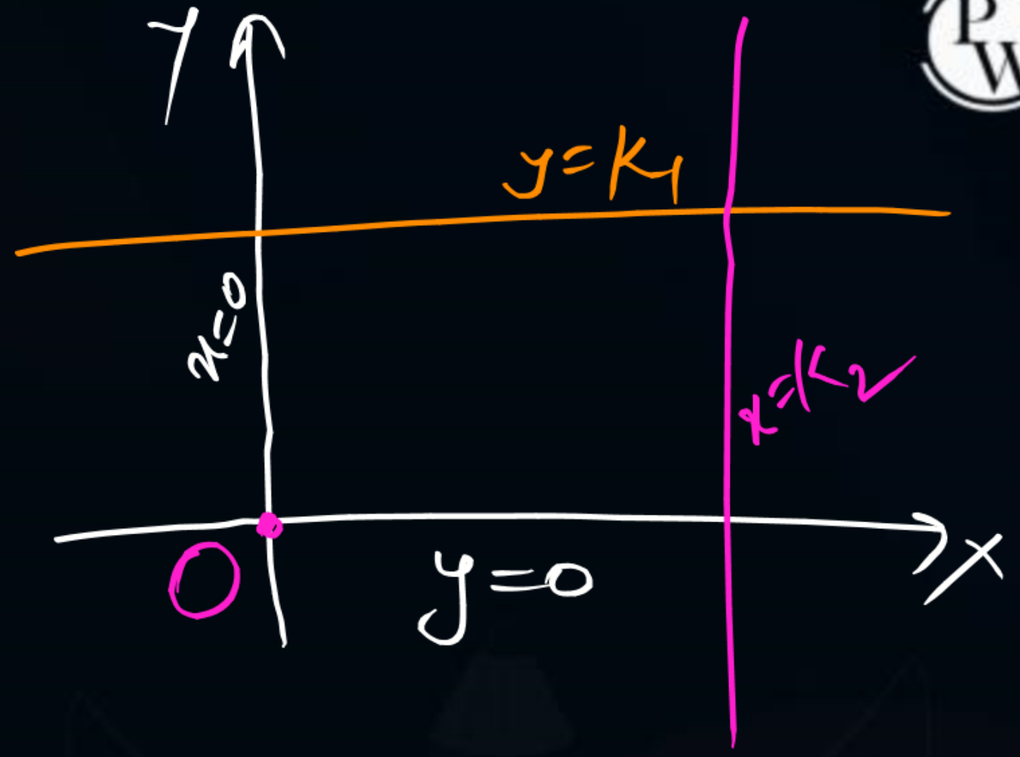
Sp Note- $y = |x| = \begin{cases} -x, & x < 0 \\ +x, & x > 0 \end{cases}$, $D_f = (-\infty, \infty)$. It's not a poly bcoz it's Dfn is not unique at all points in the Domain.

① Equ^{n} of x axis; $y = 0$

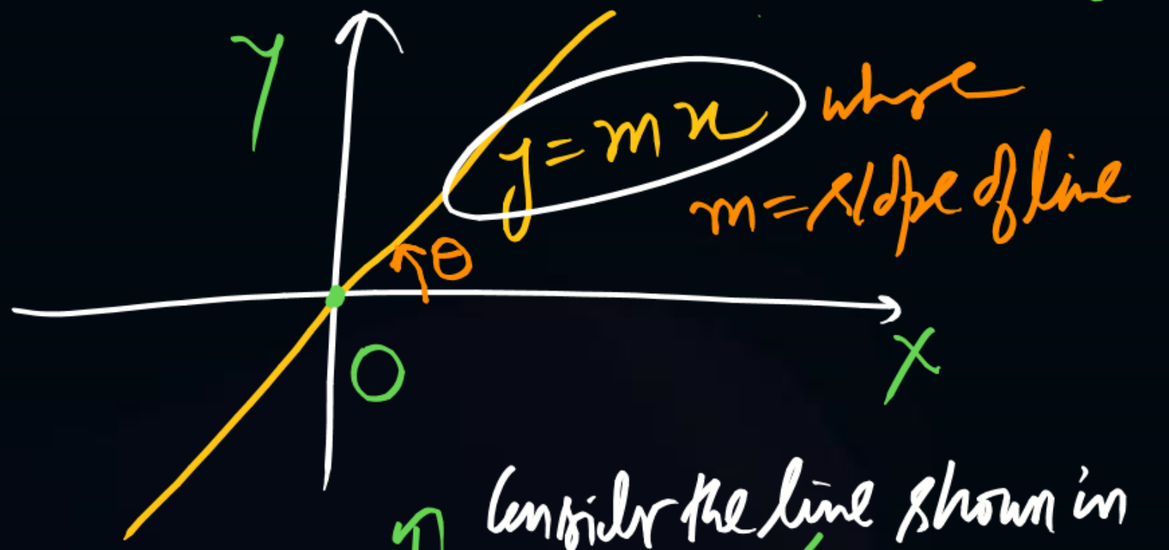
② Equ^{n} of line $\parallel^{r}$ to x axis; $y = k_1$

③ Equ^{n} of y axis; $x = 0$

④ Equ^{n} of line $\parallel^{r}$ to y axis; $x = k_2$

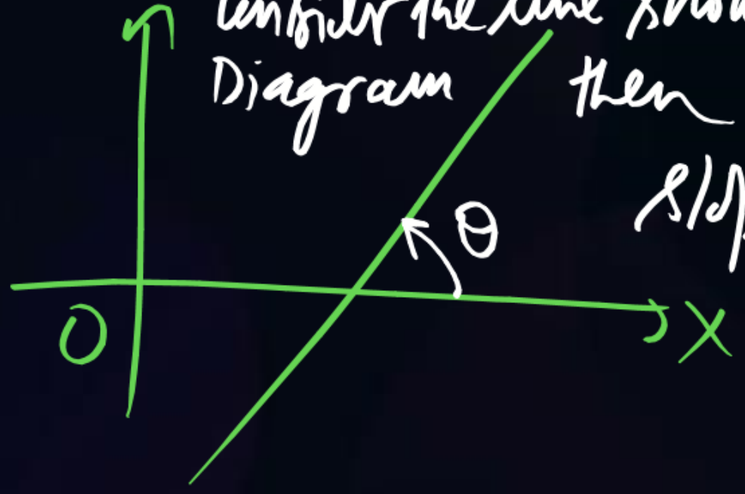


(5) Equ^n of line passing through origin



Slope:

Consider the line shown in Diagram then



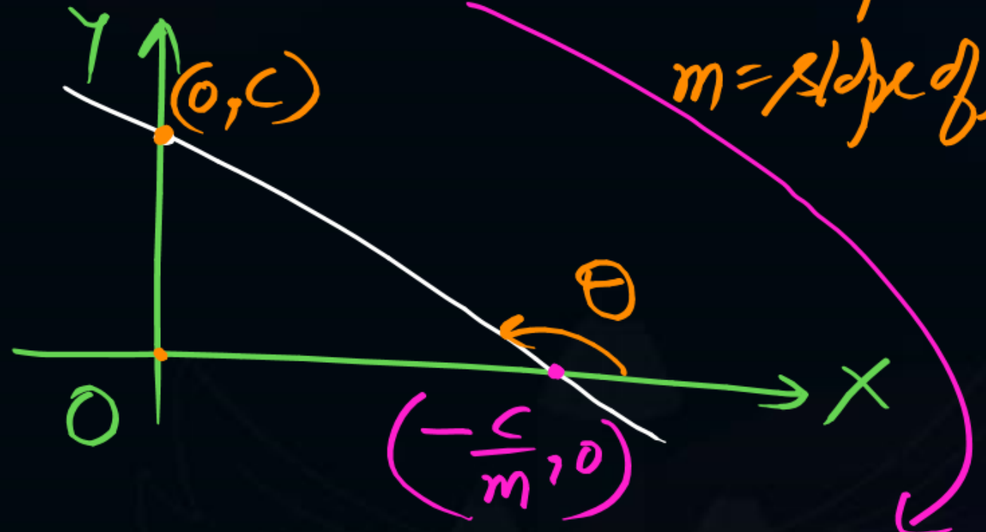
$$\text{slope of line } (m) = \tan\theta$$

(6) Slope Intercept Form of line :

$$y = mx + c$$

where c = y intercept

m = slope of line



$$-mx + y = c$$

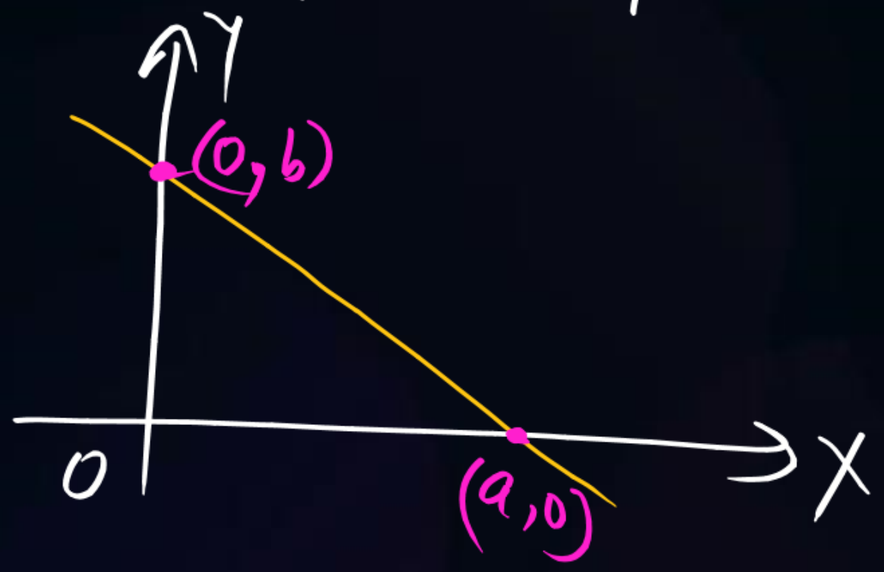
$$\frac{-mx}{c} + \frac{y}{c} = 1$$

$$\frac{x}{-c/m} + \frac{y}{c} = 1$$

(1) **Intercept form of line** →

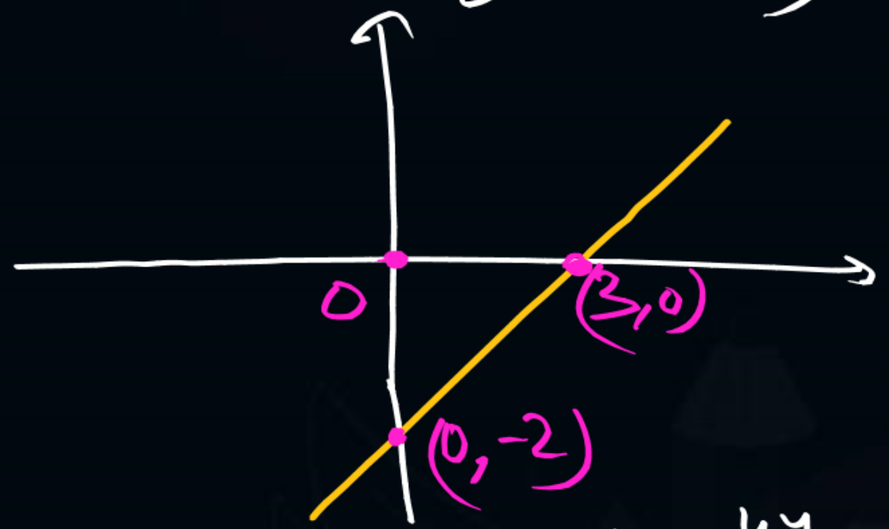
$$\frac{x}{a}+\frac{y}{b}=1$$

where a is X-Intercept
& b is Y Intercept

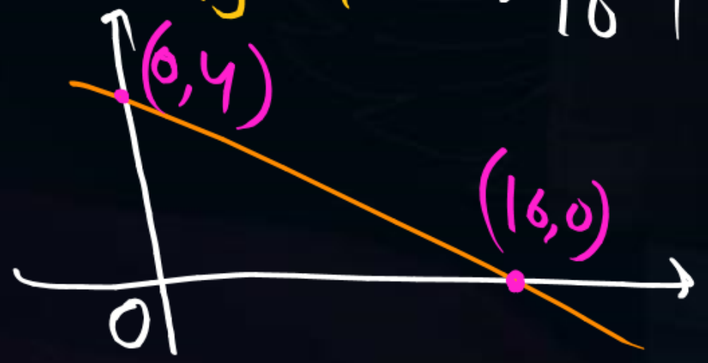


eg Draw $2x-3y=6$

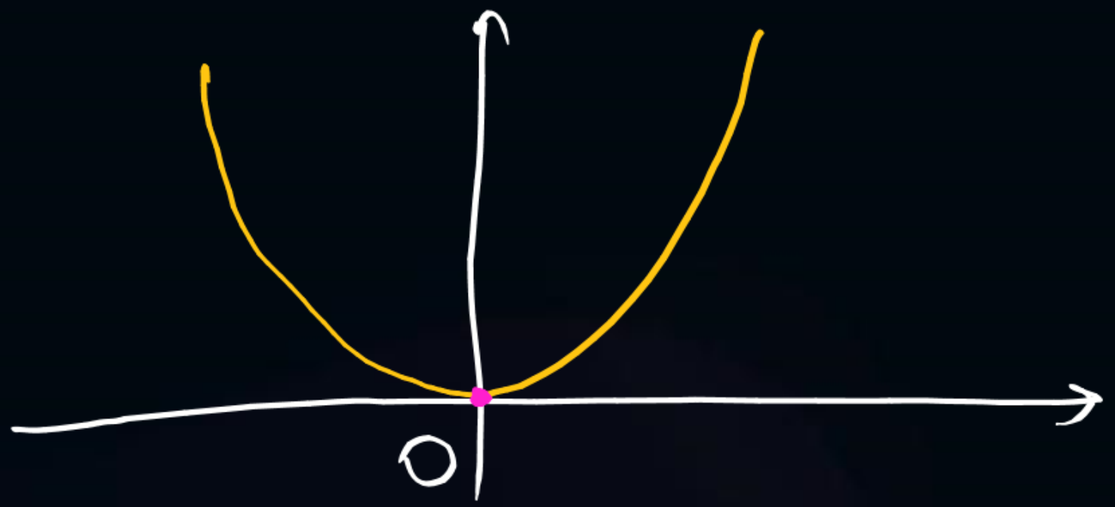
or $\frac{2x}{6}-\frac{3y}{6}=1 \Rightarrow \frac{x}{3}+\frac{y}{-2}=1$



eg $x+4y=16 \Longrightarrow \frac{x}{16}+\frac{4y}{16}=1 \Rightarrow \boxed{\frac{x}{16}+\frac{y}{4}=1}$



(8) $y = x^2, x^4, x^6, x^8 \ldots\ldots$



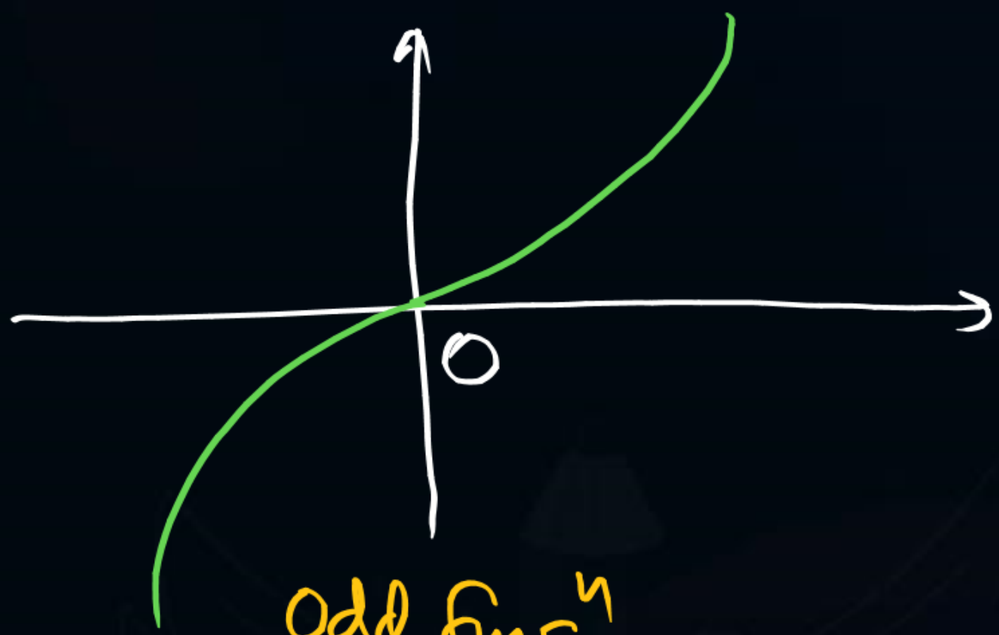
Even func^n ✓

Non Periodic

Domain : $x \in (-\infty, \infty)$

Range : $y \in [0, \infty)$

(9) $y = x^3, x^5, x^7, \ldots\ldots$



Odd func^n

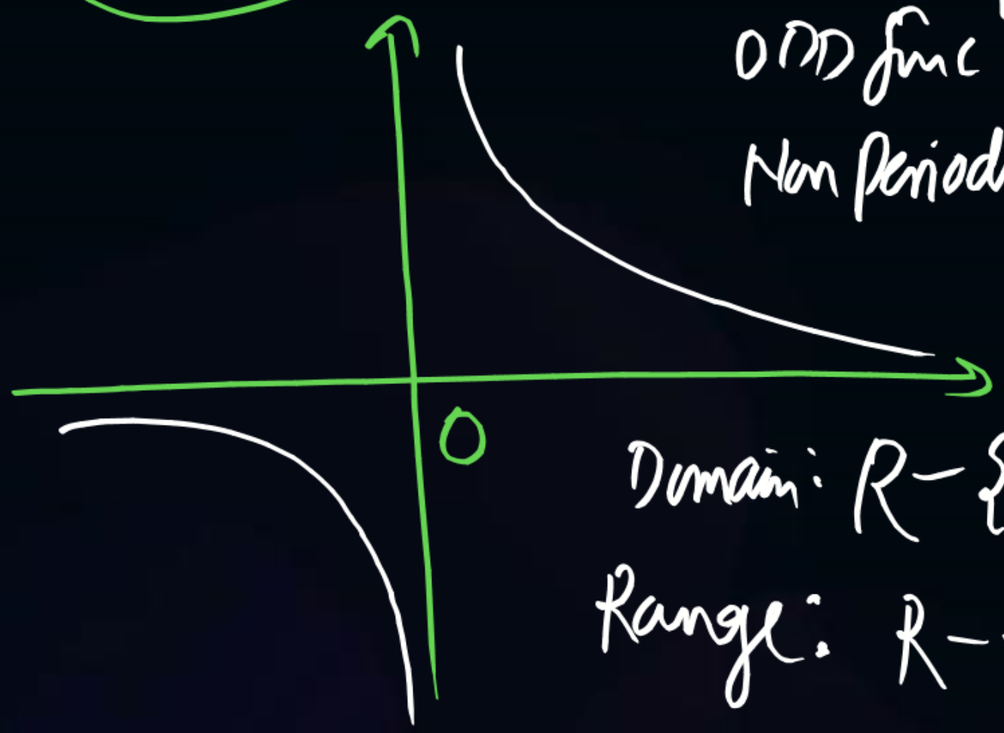
Non Periodic

Domain : $x \in (-\infty, \infty)$

Range : $y \in (-\infty, \infty)$

Rational func^n:

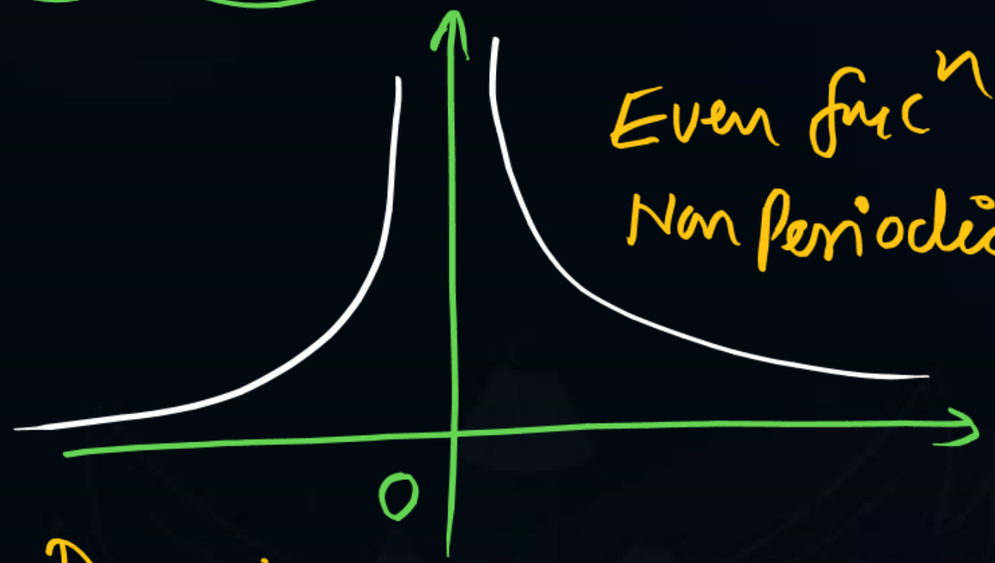
(10) $y = \frac{1}{x}, \frac{1}{x^3}, \frac{1}{x^5}, \ldots\ldots$



Odd func^n
Non Periodic

Domain: $R - \{0\}$
Range: $R - \{0\}$

(11) $y = \frac{1}{x^2}, \frac{1}{x^4}, \frac{1}{x^6}, \ldots\ldots$

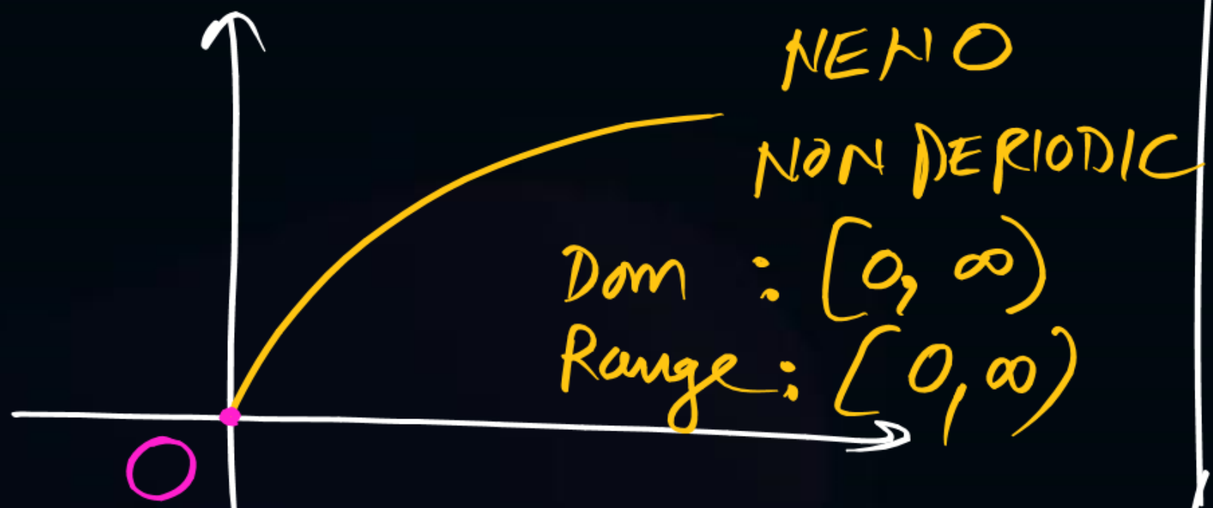


Even func^n
Non Periodic

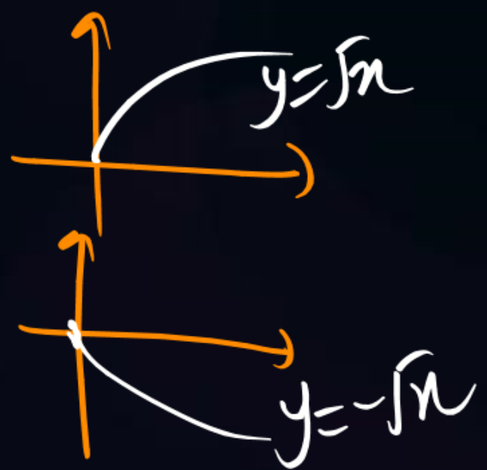
Domain : $R - \{0\}$
Range : $(0, \infty)$

IRRATIONAL func^n

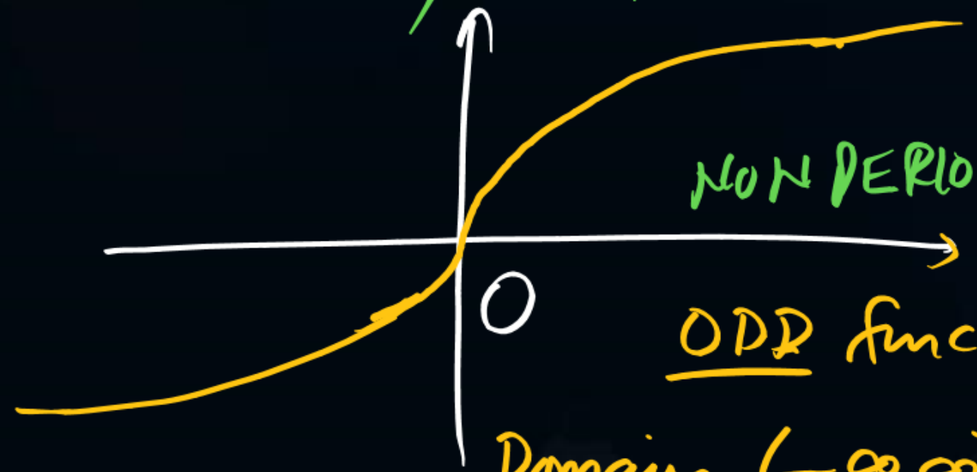
(12) $y = x^{1/2}$, $x^{1/4}$, $x^{1/6}$, ...



Note: $y^2 = x$ → $y = \sqrt{x}$, $y = -\sqrt{x}$



(13) $y = x^{1/3}$, $x^{1/5}$, $x^{1/7}$ .....



Domain $(-\infty, \infty)$

Range $(-\infty, \infty)$

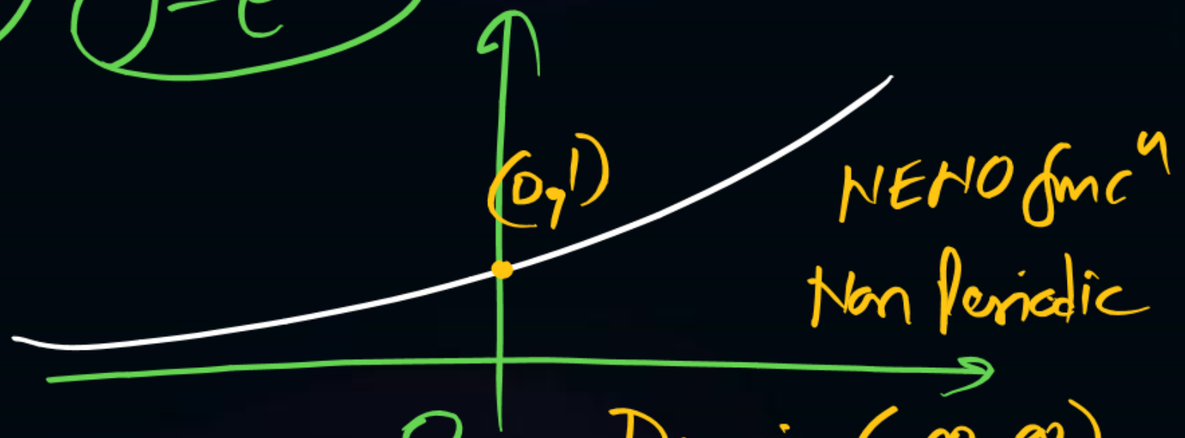
Note: $f(x) = x^{1/3}$

At $x = 1$, $f(1) = (1)^{1/3} = (1^3)^{1/3} = 1, \omega, \omega^2$

At $x = -1$, $f(-1) = (-1)^{1/3} = ((-1)^3)^{1/3} = -1, -\omega, -\omega^2$

TRANSCEDENTAL func^n.

(14) $y = e^x$



NEHO func^n

Non Periodic

Domain $(-\infty, \infty)$

Range $(0, \infty)$

$y = f(x) = e^x$

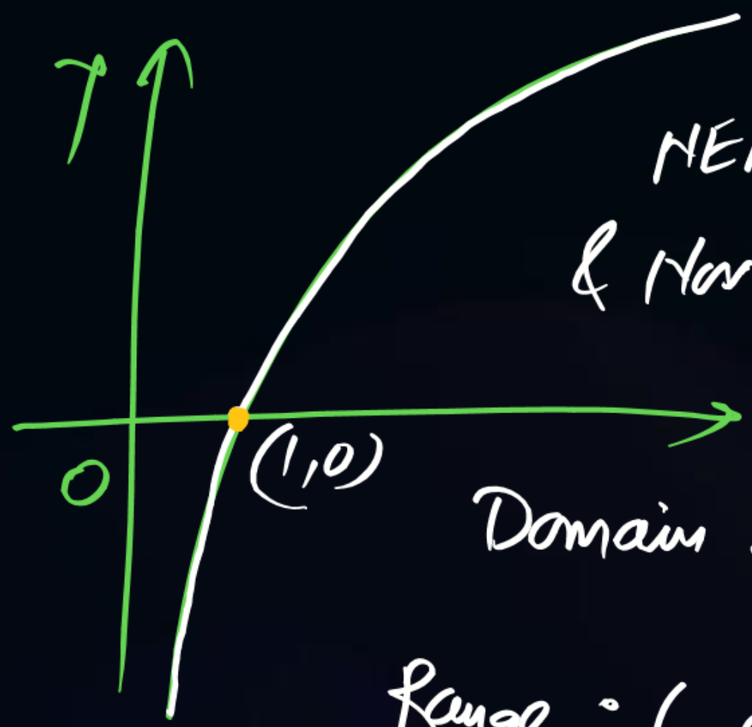
$y = f(0) = e^0 = 1$

$y = f(\infty) = e^\infty \cong \infty$

$y = f(-\infty) = e^{-\infty} \cong 0$

$\because e^{-\infty} = \frac{1}{e^\infty} = \frac{1}{\infty} \cong 0$

(16) $y = \log_e x = \ln x$



NEITHER fmc^{n} & Non periodic

Domain : $x > 0$ or $(0, \infty)$

Range : $(-\infty, \infty)$

$y = f(a) = \log_e x$

$y = \log_e 1 = 0$

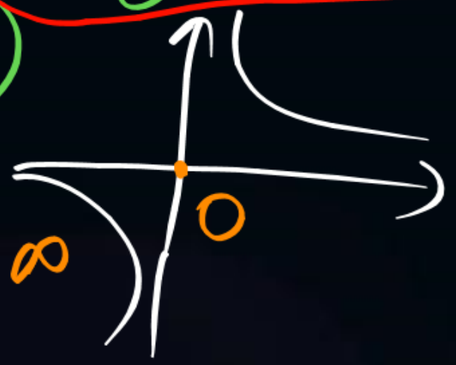
$y = \log_e(\infty) = \infty$

$y = \log_e(0) \cong -\infty$

$y = \log_e(-\infty) = N.D$

Sp. Note :- $\frac{\text{Something}}{0} = \text{N.D}$

(M-I) $y = \frac{1}{x}$



$y(0^-) = \frac{1}{0^-} = -\infty$

$y(0^+) = \frac{1}{0^+} = +\infty$

i.e. $y(0) = \text{N.D.}$

∴ In Questions, we have a liberty to take $\frac{\text{Something}}{0} = \infty$ or $\frac{\text{Something}}{0} = -\infty$



$\frac{1}{1} = 1$ $\quad$ $\frac{1}{-1} = -1$

$\frac{1}{0.1} = 10$ $\quad$ $\frac{1}{-0.1} = -10$

$\frac{1}{0.01} = 100$ $\quad$ $\frac{1}{-0.01} = -100$

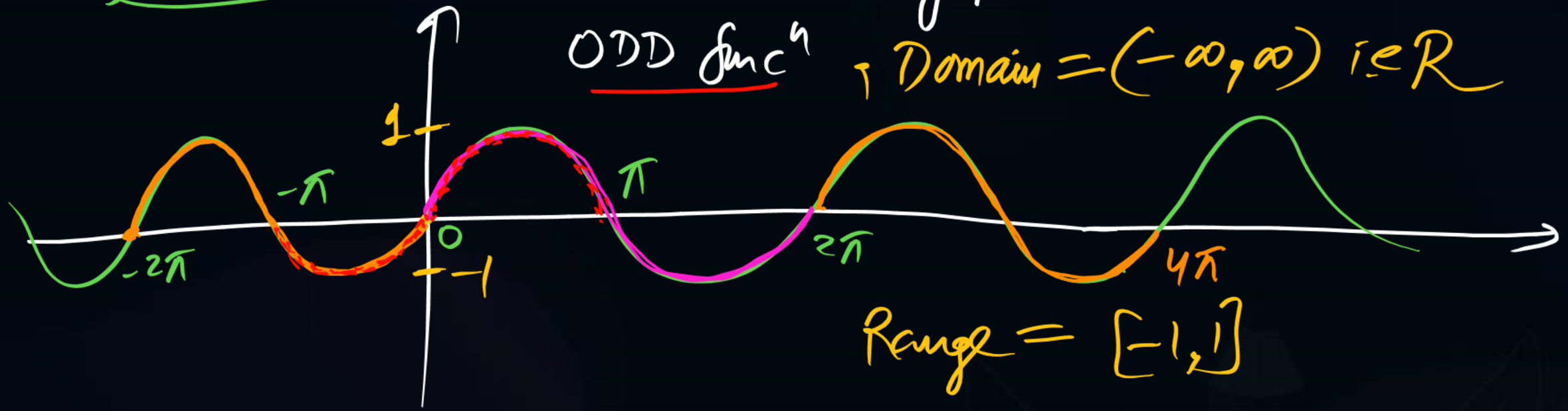
$\frac{1}{0.001} = 1000$ $\quad$ $\frac{1}{-0.001} = -1000$

$\frac{1}{0} \simeq \infty$ $\quad$ $\frac{1}{-0} = -\infty$

(17) $y = \sin x$

Periodic func^n having Period $= 2\pi$

ODD func^n , Domain $= (-\infty, \infty)$ i.e R



Range $= [-1, 1]$

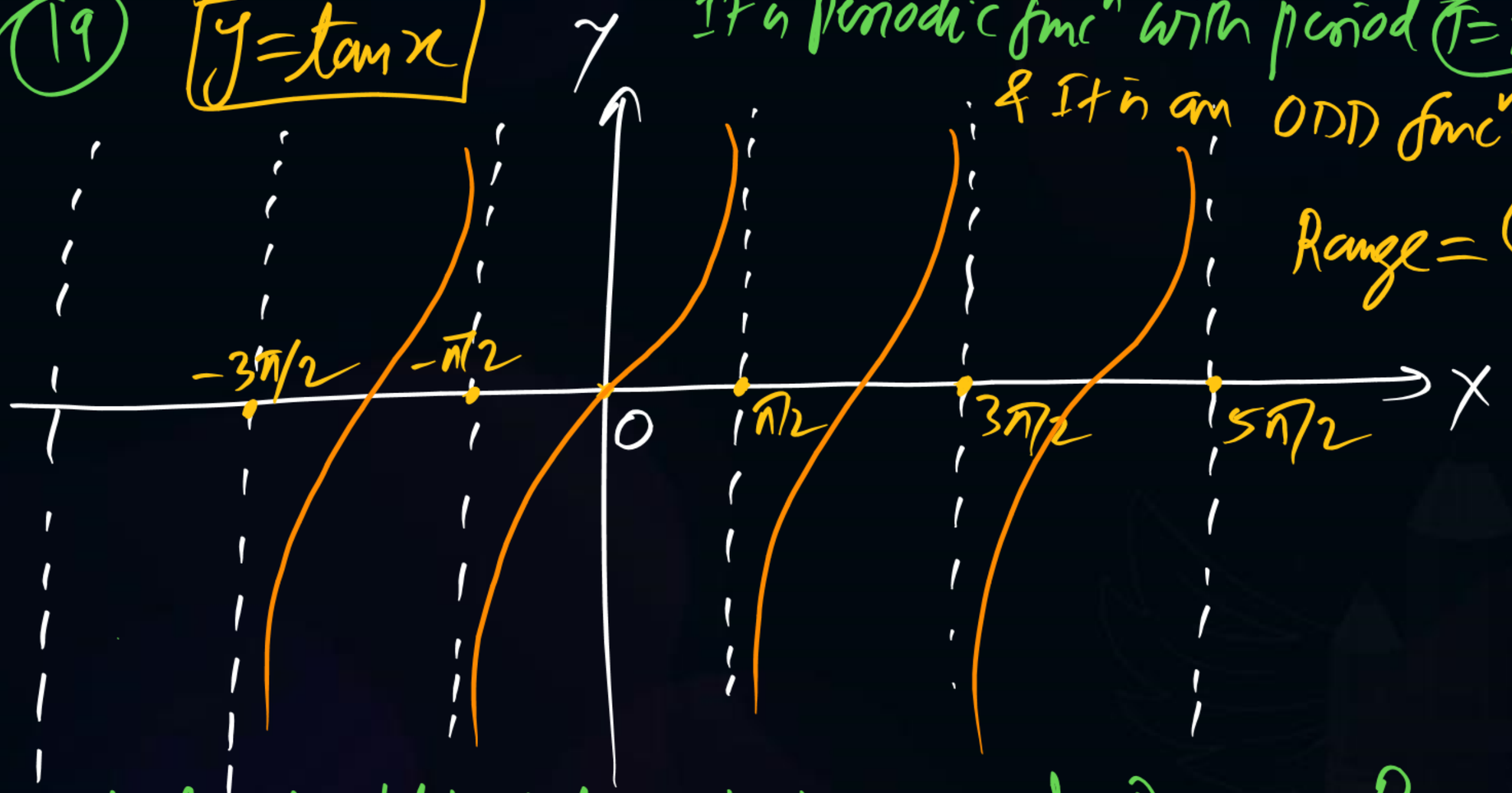
(18) $y = \cos x$, It is also periodic func^n with period $T = 2\pi$

& It is an Even func^n



Domain $= (-\infty, \infty)$ & Range $= [-1, 1]$

(19) $y = \tan x$



It is periodic fun^{n} with period $T = \pi$
& It is an ODD fun^{n}.

Range $= (-\infty, \infty)$

$y = \tan x$ is not defined at odd Multiples of $\pi/2$ $\therefore$ Domain $= R - (2n+1)\frac{\pi}{2}$

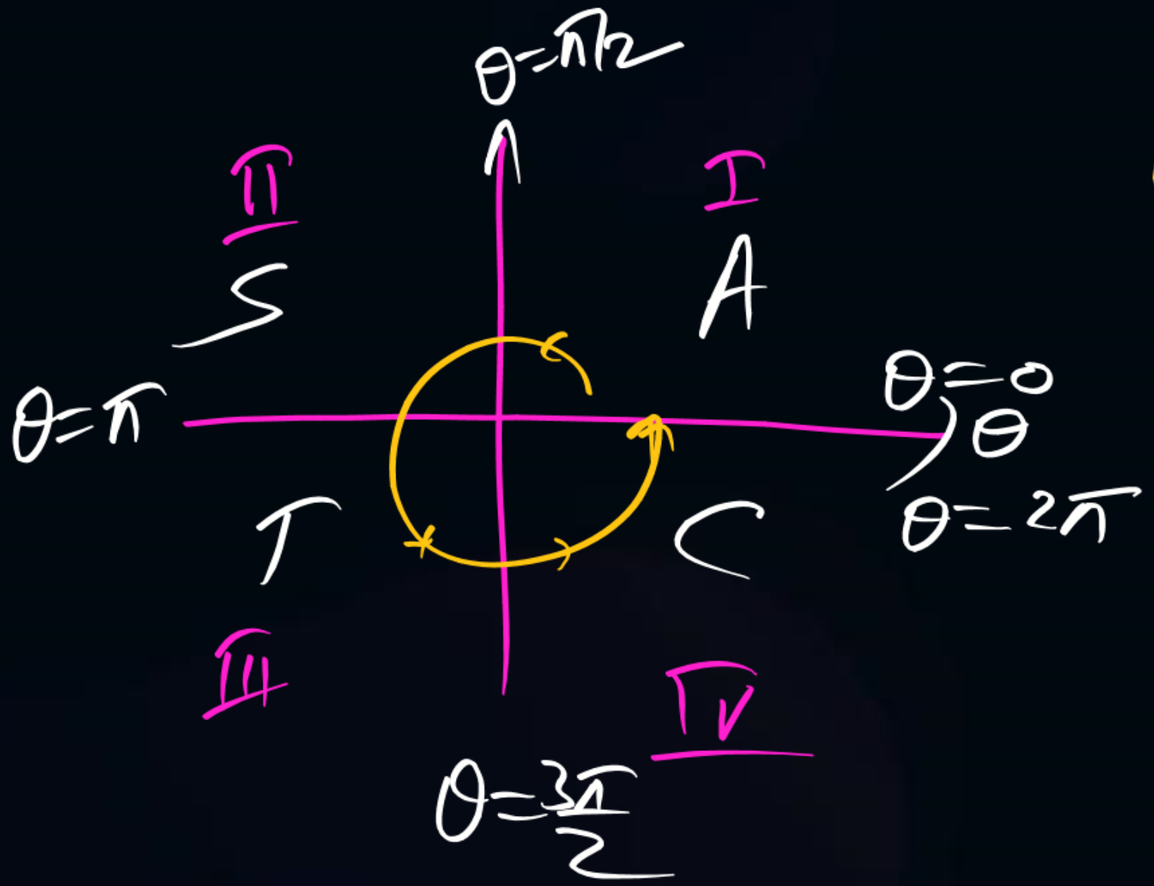
Explanation: ① $f(x) = \tan x = \frac{\sin x}{\cos x} = \frac{\text{odd func}^n}{\text{Even func}^n} = \text{odd func}^n$.

OR $f(-x) = \tan(-x) = -\tan x = -f(x) \Rightarrow$ odd func^n.

② $\tan x$ is not defined at odd Multiples of $\pi/2$ i.e.

$$\text{Domain} = R - \left\{ \ldots\ldots -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots\ldots \right\}$$

$$\text{Domain} = R - (2n+1)\frac{\pi}{2}, \quad n \in I$$



let $f(\theta)$ be an angle θ then

$\frac{\pi}{2}+\theta$ lies in II Q

$\frac{\pi}{2}-\theta$ lies in I Q

$\pi+\theta$ ∴ ; III Q

$\pi-\theta$ ∴ ; II Q

$2\pi+\theta$ ∴ ; I Q

$2\pi-\theta$ ∴ ; IV Q

$-\theta$ ∴ ; IV Q

THANK - YOU

Tel: drpuneet Sir pw